

Day 18: Introduction to Object Tracking Algorithms

Outcomes:

- Understand the difference between object detection and object tracking.
- Learn why tracking algorithms are needed after detection.
- Understand the concept of initializing a tracker with a Region of Interest (ROI).
- Know the strengths and limitations of common OpenCV trackers (KCF, CSRT, MOSSE, etc.).

From Detection to Tracking

Until now, every frame has been treated independently.

For every new frame you:

Detect motion → Detect color → Find contours → Draw a bounding box

Imagine processing video like this:

Frame 1 → Detect object

Frame 2 → Detect again

Frame 3 → Detect again

Frame 4 → Detect again

The computer keeps searching the entire image every single frame.

This works, but it is inefficient.

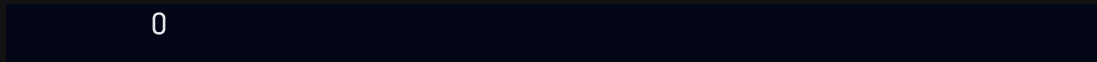
Example

Suppose you're tracking a football.

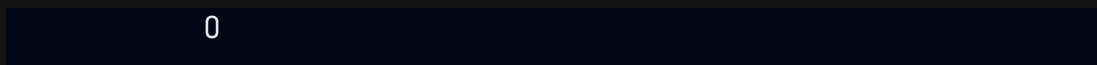
Frame 1:



Frame 2:



Frame 1:



- The ball only moved a little.
- Searching the whole frame again is unnecessary.
- Instead, we can simply follow it.

This idea is called **tracking**.

Detection vs Tracking

These two terms are often confused.

They solve different problems.

Object Detection

Detection answers: "Where is the object?"

The algorithm searches the entire image.

Example:

```
Camera Frame
+-----+
|               |
|      Object   |
|               |
+-----+
↓
Detect
↓
Bounding Box
```

Detection knows nothing about previous frames.

Every frame starts from scratch.

Object Tracking

Tracking answers: "Where did the previously known object move?"

Instead of searching the whole image, it predicts where the object has moved.

Comparison Between Detection and Tracking

Detection	Tracking
Searches whole image	Searches near previous location
Slow	Fast
Can find new objects	Cannot discover new objects
Independent every frame	Uses previous frames
Computationally expensive	Computationally cheaper

Why Detection is Expensive

Imagine a 1920×1080 video.

That contains over **2 million pixels**.

A detector may need to inspect almost every region.

Doing this 30 times every second is expensive.

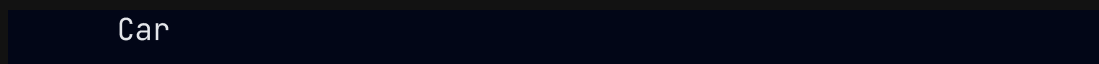
Tracking reduces the workload.

Instead of Searching every image, it performs **Search nearby**.

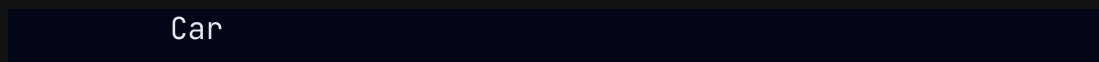
Because objects usually move smoothly.

Example:

Frame 1:



Frame 2:



The car didn't teleport.

Note: Tracking assumes motion is continuous.

Basic Tracking Pipeline

Professional computer vision systems usually work like this:



Tracker Initialization

A tracker cannot magically know what to follow.

Someone must tell it:

"Track THIS object."

This first step is called **initialization**.

Suppose your image is

```
+-----+
|                                     |
|           Object                   |
|                                     |
+-----+
```

You draw

```
+-----+
|                                     |
|   +-----+                       |
|   | Object |                       |
|   +-----+                       |
|                                     |
+-----+
```

The rectangle becomes the tracker's starting information.

The tracker stores information like:

- Size
- Shape
- Texture
- Appearance
- Position

After that, it follows the object automatically.

Region of Interest (ROI)

The rectangle used to initialize a tracker is called the **Region of Interest (ROI)**.

ROI simply means:

"The part of the image we care about."

Instead of processing everything, we only process the ROI.

```
+-----+
|                                     |
|      +-----+                    |
|      |  ROI  |                    |
|      +-----+                    |
|                                     |
+-----+
```

Tracking Starts from this ROI.

How does a Tracker Work?

Every tracker follows roughly this cycle.

Frame 1:

Initialise → Remember appearance

Frame 2:

Look nearby → Find best match → Update position

Frame 3:

Repeat

Instead of rediscovering the object, the tracker asks:

"Where does the object look most similar to last frame?"

Why Tracking alone isn't efficient

Tracking assumes the object stays visible.

Problems occur when:

- Object disappears
- Fast Motion (object moved further than expected)
- Rotation (appearance changes)
- Occlusion (temporary blocking of detected object by another object)

Hence, Detection + Tracking do the work.

Detection fixes tracker mistakes.

Tracking provides speed.

Together they are much more reliable than either alone.